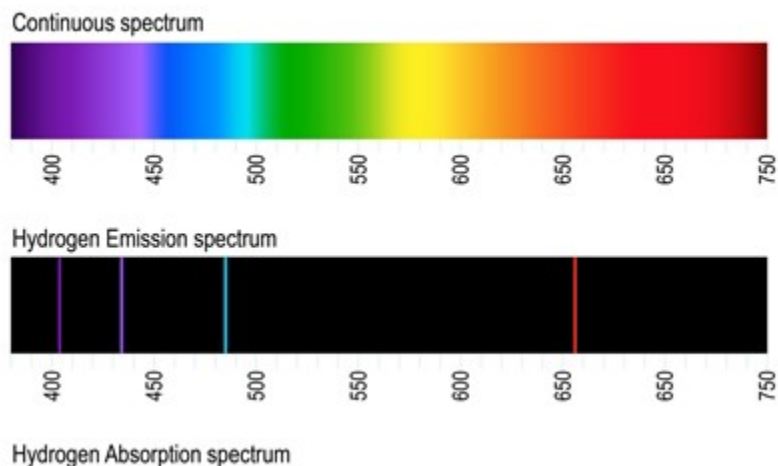


Lab 8

Spectrometry



Physical Models: Bohr Model, Hydrogen Atom, Blackbody Radiation

Analysis Tools: t-score

Experimental Systems: Hydrogen Lamp, Helium Lamp, Flashlight, Lasers

Equipment: Spectrometer, PASCO Spectrometry Software

Safety Concerns: Please be careful not to look directly into any light sources or to shine them haphazardly at other groups! Be careful handling the spectrum tubes, since they can get very hot and use a high voltage. It's recommended that you only turn them on for short periods of time to collect data, then turn them off, and to remove them by grabbing the top rather than the middle.

Introduction

In the Prelab for this week, you were introduced to the Bohr model of the atom, and worked out the predictions for the emission spectrum of hydrogen :

$$\lambda_{mn} = R^{-1} \left(\frac{1}{m^2} - \frac{1}{n^2} \right)^{-1} \quad n > m \quad (8.1)$$

Where m and n are integers which correspond to energy levels for the electron and $R = 1.09678 \times 10^7 \text{ m}^{-1}$ is a constant which characterizes the strength of the electric attraction between the electron and the nucleus.

You also learned, however, that this model is just an approximation to the full quantum theory of the hydrogen atom. In this lab, we'll test how well this model accounts of the real spectrum of hydrogen, and we'll investigate the spectrum of various sources of light more generally.

Part 1. The Spectrum of Hydrogen

Your goal in this part of the lab is to investigate the model to see how well it describes the emission spectrum of hydrogen.

Quick Check: As we always do, let's start by exploring a bit. Make sure the Spectrometer is connected to the computer via USB, and that the fiber optic cable is screwed into the spectrometer.

Log onto the lab computer and make sure to connect to the network when prompted. Go to the Finder *Applications* and open "Spectrometry". Click on "Analyze Light" at the top. On the left menu, move the top 2 scrollbars (Integration time, Number of Scans to Average) all the way to the right (you can adjust them later on if you're having trouble).

Turn on your flashlight and set it down on the table. Point your fiber optic cable at it and hit "record" in the bottom left. Wait until you get a good reading, then "stop." If you are saturating the sensor and some of your values are being capped, you can lower the "Integration time" until you can see the intensity of the whole spectrum.

This is the emission spectrum of your flashlight. Notice it is (mostly) not peaked around any particular values. This is a *continuous* spectrum.

Now try taking data using one of the red, green, or violet lasers. This may take some care in order to aim the cable carefully. You should see a single peak at the lasers' frequency. This is a *discrete* spectrum. Quantum mechanics predicts that the spectra of atoms includes (many) specific peaks like with the laser, not a continuous spectrum like the flashlight.

Now connect the hydrogen spectrum tube into the lamp. Turn the lamp on. Take note of the color. Get a reading using the spectrometer. Take a close look and make a prediction: do you think the Bohr model is likely to be accurate?

Designing Your Experiment: Now design an experiment to systematically investigate whether or not the Bohr model is a good explanation for the emission lines you record. You should:

- Record all of the emission lines you see in the spectrometer data.
- Decide on what the uncertainty in each line is.
- Record all values of the wavelength that are predicted by the model which fall within the spectrometer range of 350 to 1000 nm.
- Use 1nm as the uncertainty in the predicted wavelengths of the Bohr model.
- Decide on a criterion for assessing the model based on comparing predicted vs. observed spectral lines. Think about the statistical tools you have learned in this class.
- What if there are more or fewer lines than predicted? How will you handle this?

Recall that Prelab 8 included help with delineating all of the predicted values of the wavelength.

Conducting Your Experiment: Go ahead and give it a shot. You can change your method around while you're at it. Just make sure to record any changes.

Forming a Conclusion: Did the model and data match exactly? Did it match for some lines and not for others? Are there any patterns? Make an assessment. Remember that deviations from the model could relate to aspects of hydrogen or quantum mechanics not accounted for by the Bohr model.

Check in with Another Group: : Check back in with the same group you discussed things with before. How do your results compare? Note two similarities and/or differences between what you found, and use this to inform your proposals for future iterations.

A final question: The spectrum of hydrogen can be divided into *spectral series*:

Named Series	Transitions	Radiation Type
Lyman	$n \rightarrow 1$	Ultraviolet
Balmer	$n \rightarrow 2$	Visible
Paschen	$n \rightarrow 3$	Infrared

Is your observations consistent with the descriptions in the table? Which series is predominant in your data? Have you observed transitions that are not part of the series?

Part 2: Spectra in General

In this part of the lab, you're free to explore the spectra of other objects. Here are some examples for you to consider:

- How does intensity depend on distance? Define a measure of intensity and find out if it decreases with distances as $1/r^2$ for the flashlight and a light source.
- How does the spectrum of light change due to transmission? For example measure the spectrum of a light source with and without a color filter. You can find a transition coefficient for different wavelengths $T = I_{transmitted}(\lambda)/I_{source}(\lambda)$.

But you're welcome to offer your own as well, just check with your TA before proceeding.

Check in with Another Group: Briefly discuss your plans with another group. They may provide you with additional ideas and feedback. Note one similarity and one difference between your approaches.

Design your experiment, **Conduct** it as planned, and **Form a Conclusion:** all like you usually would.

Check in with Another Group: Check back in with the same group you discussed things with before. How do your results compare? Note two similarities and/or differences between what you found, and use this to inform your proposals for future iterations.

Writing Up Your Lab Notes for Submission

Write up your notes for Part 1 and Part 2 in an organized way according to the rubric provided.